

LAB 1: Plots from known probability distributions

Bayesian Statistics - a.y. 2026/2027

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Gamma distribution

The probability density function (pdf) of a Gamma random variable is

$$f(x) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} \exp(-\beta x) \mathbb{I}_{[x \geq 0]}$$

with $\alpha, \beta > 0$. How does (α, β) influence the shape of this distribution?

```
gamma_pdf <- function(x, alpha, beta) {  
  beta^alpha / gamma(alpha) *  
    (x^(alpha - 1)) *  
    exp(-beta * x) *  
    as.numeric(x >= 0)  
}
```

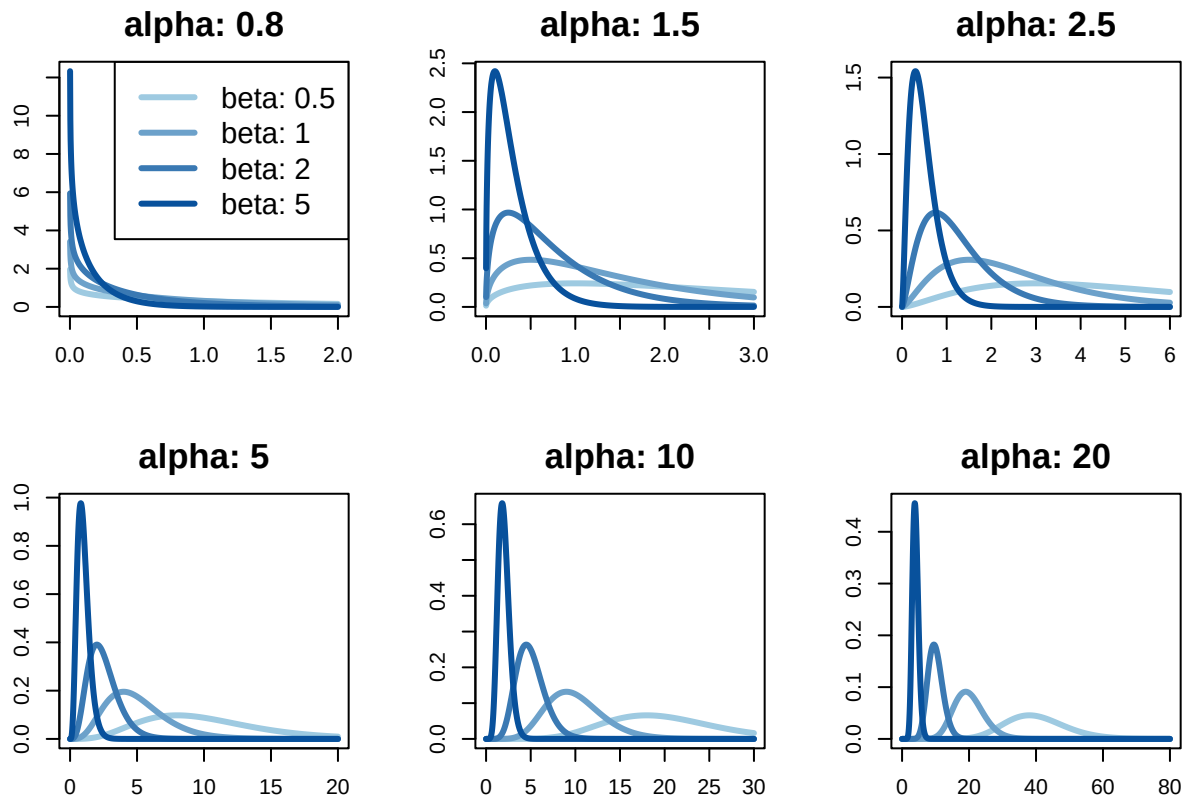
```
par(mfrow = c(2, 3), mar = c(4, 4, 3, 1))  
  
alphas <- c(0.8, 1.5, 2.5, 5, 10, 20)  
betas <- c(0.5, 1, 2, 5)  
  
xgrids <- list(  
  seq(0.001, 2.0, length.out = 1000),  
  seq(0.001, 3.0, length.out = 1000),  
  seq(0.001, 6.0, length.out = 1000),  
  seq(0.001, 20, length.out = 1000),  
  seq(0.001, 30, length.out = 1000),  
  seq(0.001, 80, length.out = 1000)  
)  
  
colormap <- colorRampPalette(c("#9ECAE1", "#08519C"))(length(betas))  
  
for (i in seq_along(alphas)) {  
  alpha <- alphas[i]  
  
  y_vals <- sapply(betas, function(b) gamma_pdf(xgrids[[i]], alpha, b))  
  
  plot(xgrids[[i]], y_vals[, 1], type = "l", col = colormap[1], lwd = 3,  
    ylim = c(0, max(y_vals)), xlab = "", ylab = "",  
    main = sprintf("alpha: %g", alpha), cex.main = 1.6)  
  
  for (j in 2:length(betas)) {
```

```

    lines(xgrids[[i]], y_vals[, j], col = colormap[j], lwd = 3)
  }

  if (i == 1) {
    legend("topright", legend = sprintf("beta: %g", betas),
          col = colormap, lwd = 3, cex = 1.4)
  }
}

```



Beta distribution

The pdf of a Beta random variable is the following:

$$f(x) = \frac{1}{B(\alpha_1, \alpha_2)} x^{\alpha_1-1} (1-x)^{\alpha_2-1} \mathbb{I}_{[x \in (0,1)]}$$

with $\alpha_1, \alpha_2 > 0$.

```

beta_fn <- function(alpha_1, alpha_2) {
  gamma(alpha_1) * gamma(alpha_2) / gamma(alpha_1 + alpha_2)
}

beta_pdf <- function(x, alpha_1, alpha_2) {
  x^(alpha_1 - 1) * (1 - x)^(alpha_2 - 1) / beta_fn(alpha_1, alpha_2) *

```

```

    as.numeric(x > 0) * as.numeric(x < 1)
}

par(mfrow = c(2, 3), mar = c(4, 4, 3, 1))

alphas_1 <- c(0.5, 1, 2, 5, 10, 20)
alphas_2 <- c(0.5, 1, 2)

xgrid <- seq(0.001, 0.999, length.out = 1000)

colormap <- hcl.colors(length(alphas_2), palette = "Viridis", rev = TRUE)

for (i in seq_along(alphas_1)) {
  alpha_1 <- alphas_1[i]

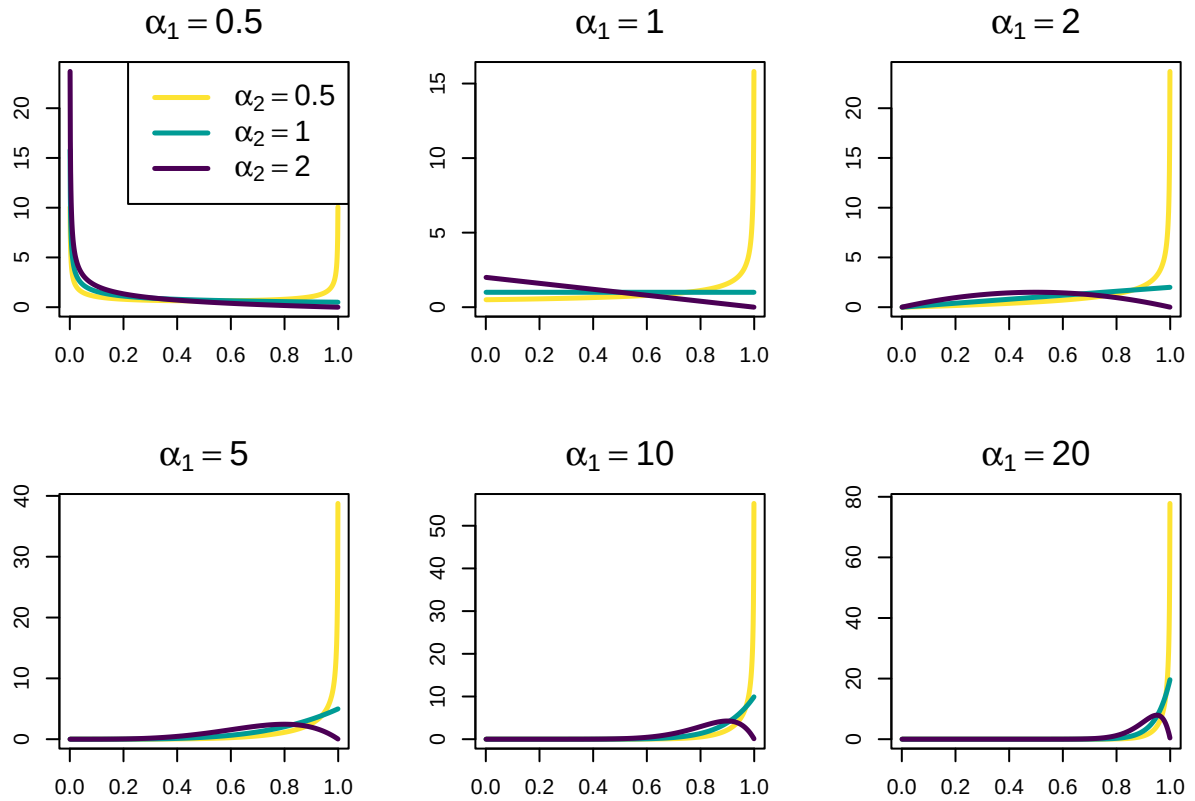
  y_vals <- sapply(alphas_2, function(b) beta_pdf(xgrid, alpha_1, b))

  plot(xgrid, y_vals[, 1], type = "l", col = colormap[1], lwd = 2.5,
       ylim = c(0, max(y_vals)), xlab = "", ylab = "",
       main = bquote(alpha[1] == .(alpha_1)), cex.main = 1.6)

  for (j in 2:length(alphas_2)) {
    lines(xgrid, y_vals[, j], col = colormap[j], lwd = 2.5)
  }

  if (i == 1) {
    legend_labels <- sapply(alphas_2, function(a2) bquote(alpha[2] == .(a2)))
    legend("topright", legend = as.expression(legend_labels),
          col = colormap, lwd = 2.5, cex = 1.4)
  }
}

```



Dirichlet distribution

The pdf of the Dirichlet random variable is:

$$f(x) = \frac{1}{B(\alpha_1, \dots, \alpha_k)} \prod_{j=1}^k x_j^{\alpha_j-1} \mathbb{I}_{[x \in S^{k-1}]}$$

The pdf is highly influenced by the value of $\alpha_1, \dots, \alpha_k$: higher values of α_i relative to the other parameters will increase the likelihood of observing a higher value for the corresponding x_i . Here we use the R package **ternary** to visualize this on the 2-dimensional simplex.

```
multi_beta_fn <- function(alphas) {
  prod(gamma(alphas)) / gamma(sum(alphas))
}

# install the package if needed
# install.packages("Ternary")
library(Ternary)

alphas <- list(
  rep(1, 3),
  c(0.9, 0.9, 0.9),
  c(5, 5, 5),
  c(2, 4, 8)
```

```

)

par(mfrow = c(2, 2), mar = c(1, 1, 1, 1))

my_spectrum <- hcl.colors(256, palette = "YlOrRd")

for (i in 1:4) {
  current_alphas <- paste(alphas[[i]], collapse = ", ")
  plot_title <- paste0("Alpha = (", current_alphas, ")")

  TernaryPlot(
    alab = "x1",
    blab = "x2",
    clab = "x3",
    axis.labels = c(0, NA, 0.2, NA, 0.4, NA, 0.6, NA, 0.8, NA, 1),
    main = plot_title,
    lab.cex = 0.8,
    axis.cex = 0.6,
    cex = 0.8
  )

  dirichlet_pdf <- function(a, b, c) {
    # a -> x1 , b -> x2 , c -> x3
    alp <- alphas[[i]]

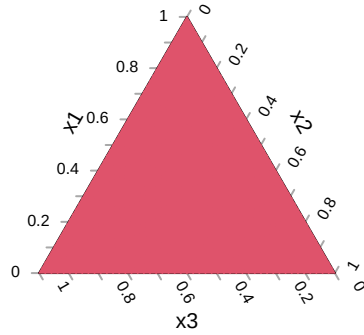
    (a^(alp[1] - 1) * b^(alp[2] - 1) * c^(alp[3] - 1)) / multi_beta_fn(alp)
  }

  cols <- TernaryPointValues(dirichlet_pdf)

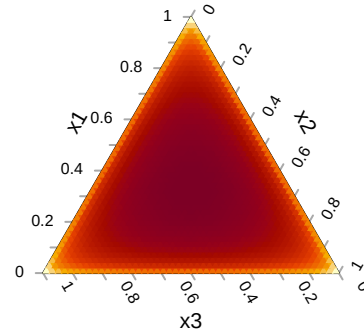
  if (min(cols[3, ], na.rm = TRUE) == max(cols[3, ], na.rm = TRUE)) {
    values <- ColorTernary(cols, filled = TRUE, spectrum = NULL, resolution = 36L)
  } else {
    values <- ColorTernary(cols, filled = TRUE, spectrum = my_spectrum, resolution = 36L)
  }
}

```

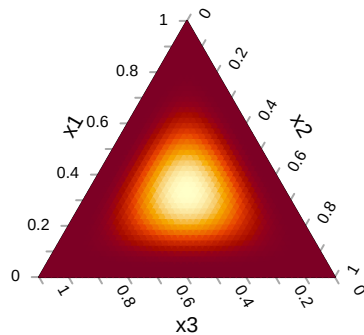
Alpha = (1, 1, 1)



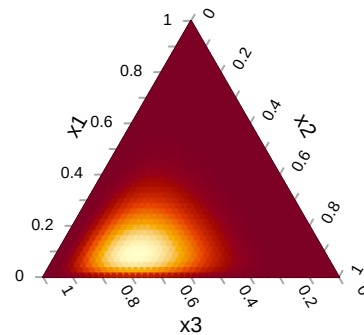
Alpha = (0.9, 0.9, 0.9)



Alpha = (5, 5, 5)



Alpha = (2, 4, 8)



Inverse Gamma distribution

Recall that the pdf of the Inverse Gamma distribution is given by:

$$f(x) = \frac{\beta^\alpha}{\Gamma(\alpha)} \left(\frac{1}{x} \right)^{\alpha+1} e^{-\frac{\beta}{x}} \mathbb{I}_{(0,+\infty)}(x)$$

```
xgrid <- seq(0.01, 10, length.out = 1000)

dinvgamma <- function(x, shape, scale) {
  (scale^shape / gamma(shape)) * (1 / x)^(shape + 1) * exp(-scale / x)
}

y <- dinvgamma(xgrid, shape = 10.0, scale = 10.0)

plot(xgrid, y, type = "l", col = "black", lwd = 1, xlab = 'x')
```

